

Learning Objective 1.4: I can calculate input impedance, feed considerations, and the role of baluns in an antenna feed system.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Radiation resistance and efficiency.** A short dipole has length $\ell = 0.08\lambda$. Its conductors contribute a loss resistance of $R_{\text{loss}} = 1.5 \, \Omega$.

(a) Compute the radiation resistance $R_{\text{rad}} = 80\pi^2(\ell/\lambda)^2$.

$R_{\text{rad}} =$

(b) Compute the radiation efficiency η_{rad} .

$\eta_{\text{rad}} =$

(c) The short dipole's directivity is $D = 1.5$ (1.76 dBi).
Find the gain in dBi.

$G =$

2. Feeding a half-wave dipole with $50\ \Omega$ coax.

- (a) A dipole trimmed to resonance presents $Z_{\text{in}} = 70 + j0\ \Omega$. Find Γ , VSWR, return loss, and mismatch loss on a $50\ \Omega$ line.

 $\Gamma =$

VSWR =

RL =

 $L_{\text{mm}} =$

- (b) An *untrimmed* half-wave dipole presents $Z_{\text{in}} = 73 + j42.5\ \Omega$. Find $|\Gamma|$ and VSWR on the same line.

 $|\Gamma| =$

VSWR =

- (c) Compare (a) and (b): in one sentence, what did trimming to resonance buy you, and why?

- (d) A third dipole on the same $50\ \Omega$ line measures VSWR = 1.5, and you know it is purely resistive at that frequency. What two real Z_{in} are consistent with that reading? State what a VSWR measurement alone can and cannot tell you.

 $Z_{\text{in}} =$

3. **Quarter-wave transformer.** A resonant antenna presents a real input impedance $R_L = 36 \, \Omega$. You want to match it to a $50 \, \Omega$ feed line with a quarter-wave transformer.

- (a) What characteristic impedance Z_1 must the $\lambda/4$ section have?

$Z_1 =$

- (b) What is Γ at the design frequency?

- (c) In one sentence, what happens to the match off the design frequency, and why?

- (d) The antenna in this problem is purely resistive. Explain in one or two sentences why a quarter-wave transformer is the wrong tool the moment Z_{in} has a reactive part.

4. **L-match design.** A compact antenna presents $Z_{\text{in}} = 25 - j40 \, \Omega$ at $f = 1.5\text{GHz}$. You must match it to a $50 \, \Omega$ line with lumped elements. Work the match in two moves: *cancel the reactance*, then *transform the resistance*.

- (a) Find $|\Gamma|$ and VSWR for the raw, unmatched antenna.

$|\Gamma| =$

$\text{VSWR} =$

- (b) **Move 1.** What series reactance cancels the load reactance, and what component value does it correspond to at 1.5GHz?

$X_1 =$

$L_1 =$

- (c) **Move 2.** With $R_L = 25 \, \Omega$ real, design the L-network that transforms it to $50 \, \Omega$. Use $Q = \sqrt{Z_0/R_L - 1}$, series $X_s = QR_L$ on the load side and shunt $X_p = Z_0/Q$ on the line side. Give both reactances and both component values, and combine the two series inductors.

$L_{\text{tot}} =$

$C =$

- (d) The loaded Q of this match came out at 1. Compare the bandwidth and the practicality of this L-match against the $\lambda/4$ transformer of Question 3, and say which you would build for a 1.5GHz handheld radio.

5. **Mismatch that kills the amplifier.** A power amplifier (PA) delivers a forward power of $P_{\text{fwd}} = +40$ dBm toward a transmit antenna. A shunt protection diode at the PA output fails catastrophically if it must dissipate more than $P_{\text{max}} = +30$ dBm. Model *all* of the reflected power as landing on the diode, so $P_{\text{ref}} = |\Gamma|^2 P_{\text{fwd}}$.

(a) Convert P_{fwd} and P_{max} to watts.

$$P_{\text{fwd}} = \boxed{}$$

$$P_{\text{max}} = \boxed{}$$

(b) Find the largest $|\Gamma|$ the system can tolerate before the diode is destroyed.

$$|\Gamma|_{\text{max}} = \boxed{}$$

(c) Convert to a VSWR limit and a return-loss limit, and state the datasheet rule that follows.

$$\begin{array}{l} \text{VSWR}_{\text{max}} = \boxed{} \\ \text{RL}_{\text{min}} = \boxed{} \end{array}$$

(d) In one sentence: how does a circulator or isolator between the PA and antenna change this picture?

6. Baluns.

- (a) In one or two sentences, explain why feeding a dipole *directly* with coax (no balun) distorts the pattern.

- (b) You must feed a folded dipole ($Z_{\text{in}} \approx 292 \, \Omega$) with $75 \, \Omega$ coax. Which balun would you choose, and what does it do?

- (c) You feed a straight resonant dipole ($\approx 70 \, \Omega$) with $50 \, \Omega$ coax. Which balun would you choose, and why is it a *different* choice than (b)?

Documentation: _____